

ToolCall M60 scaling study

Fixed 60.4M-parameter model · comparison across training-token budgets

COMPLETED

3/3 runs

BEST GENERAL LOSS

3.3628 · High · 100M

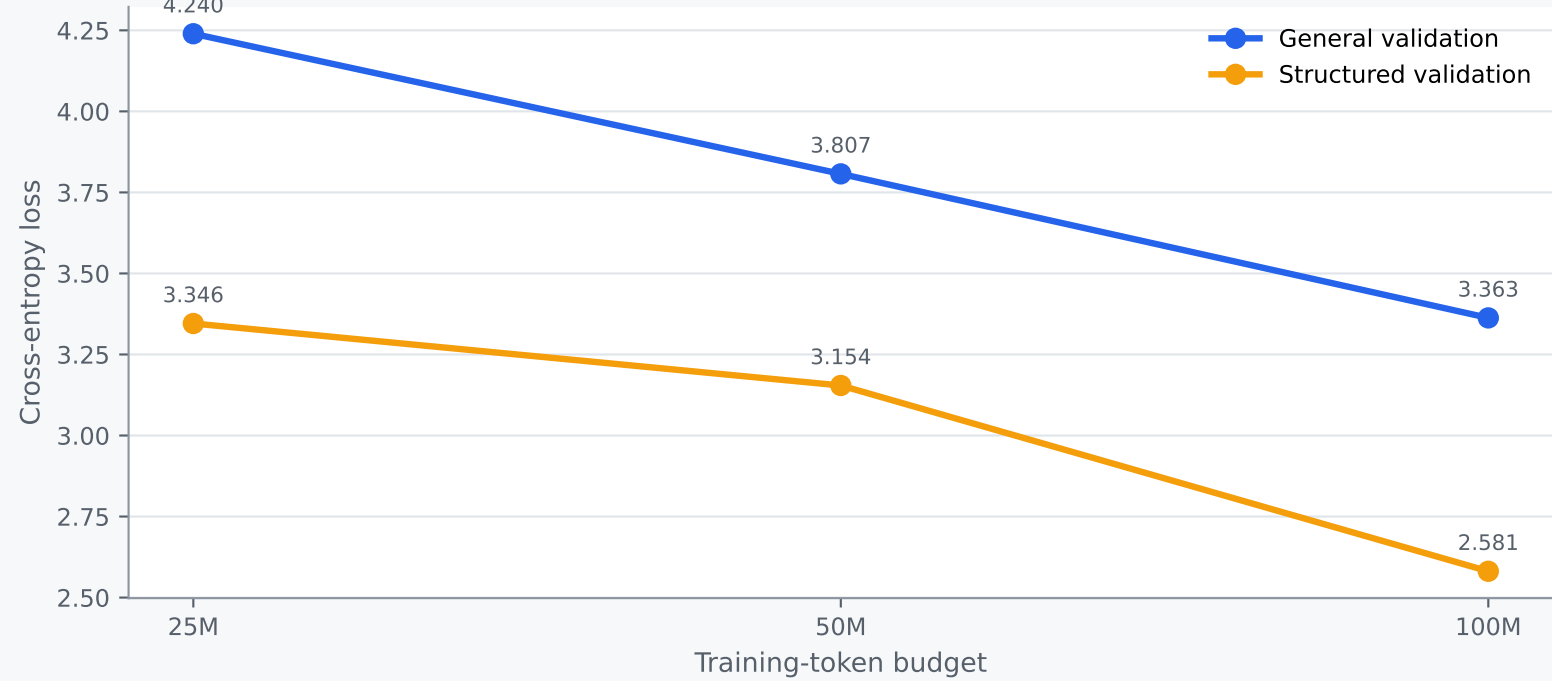
BEST STRUCTURED LOSS

2.5812 · High · 100M

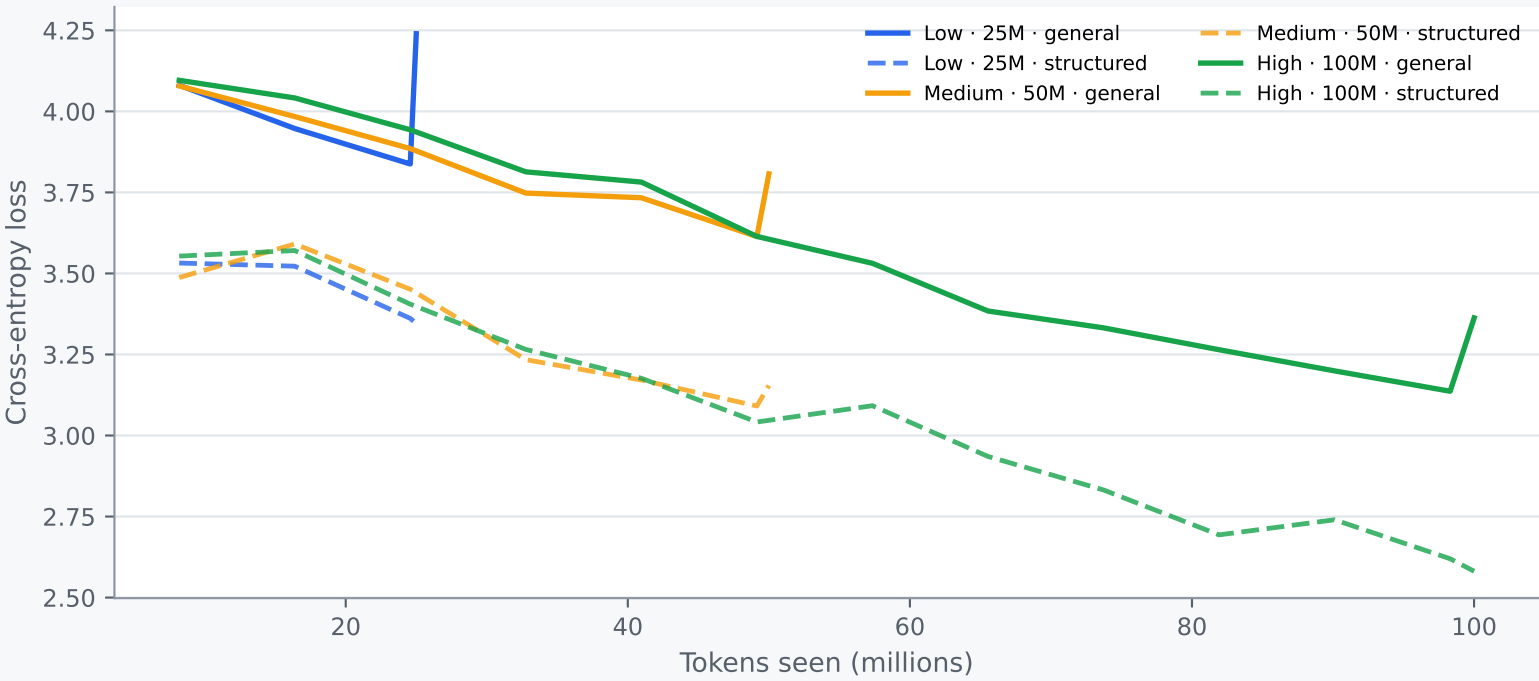
TOTAL GPU TIME

2.20 hours

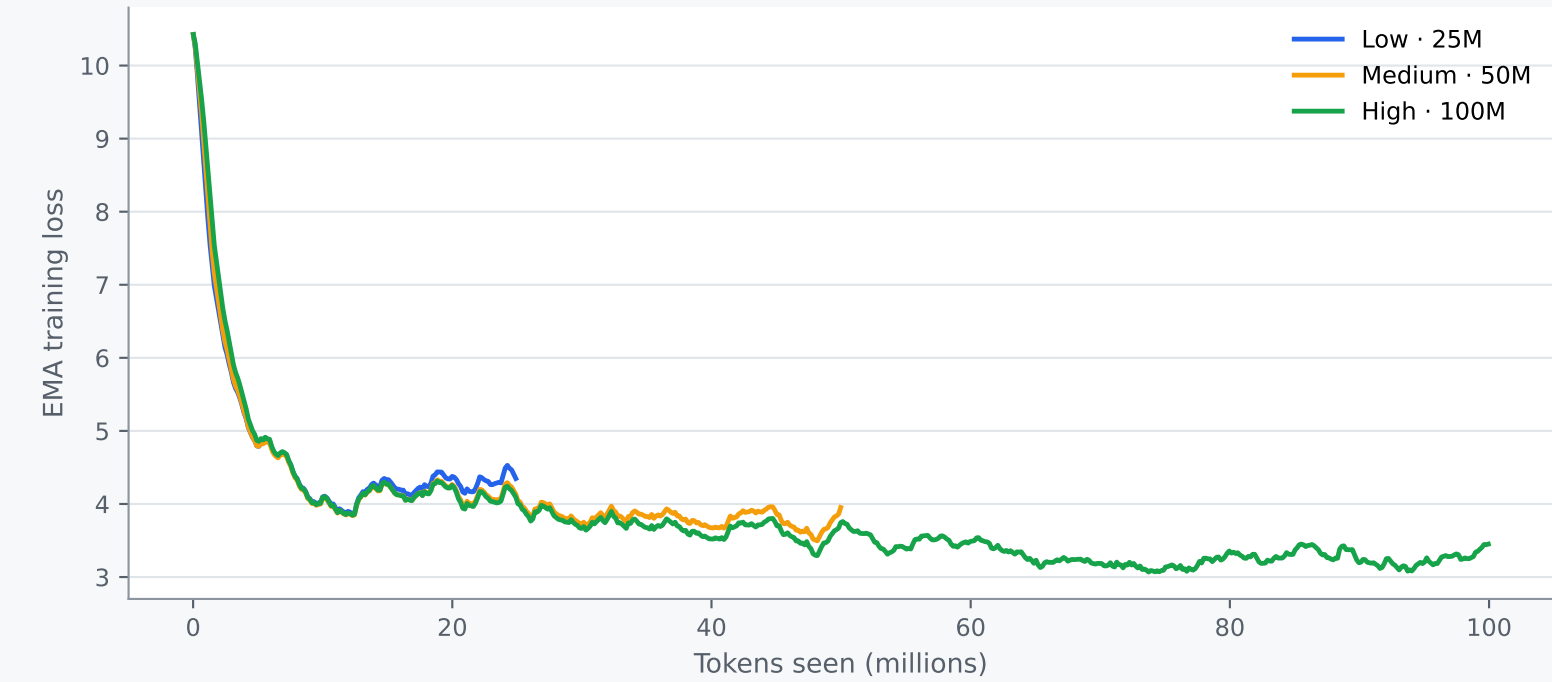
Final validation loss vs token budget



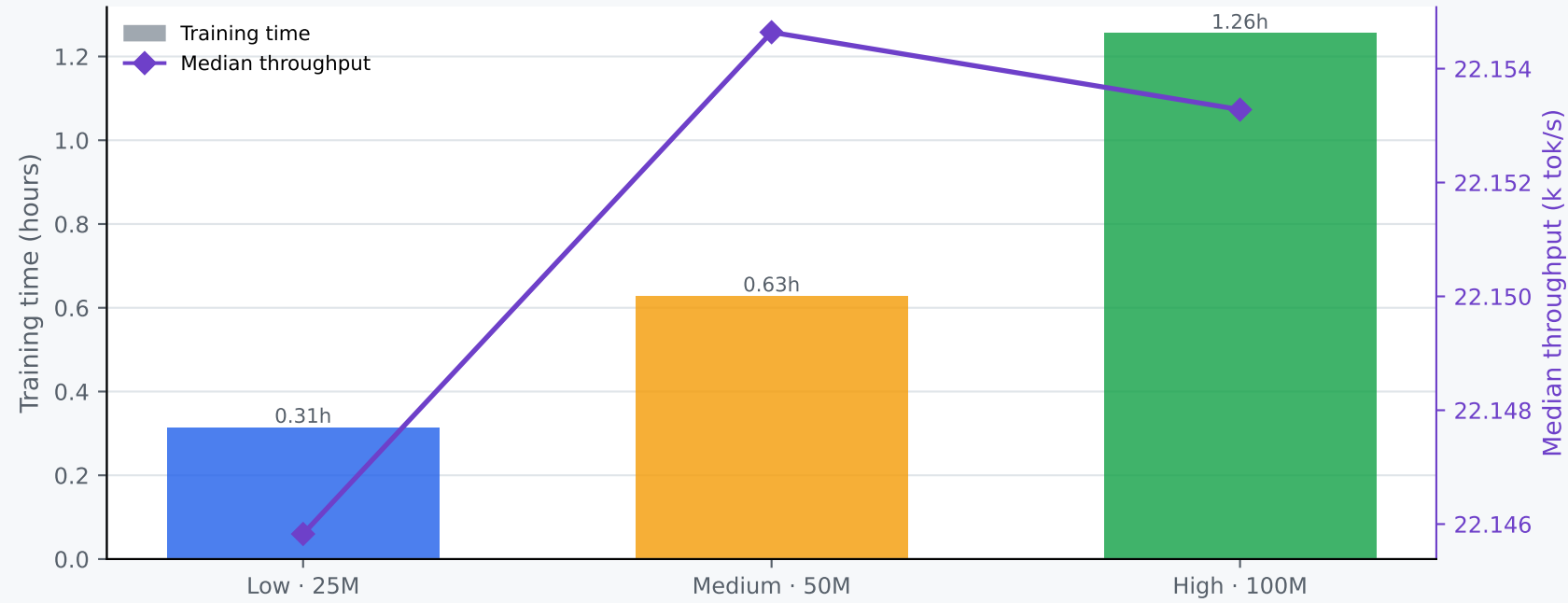
Validation learning curves



Training-loss convergence



Runtime and throughput



Run comparison

Run	Status	Tokens	General loss	Structured loss	Train time	Median speed	Peak VRAM
Low · 25M	completed	25.0M	4.2395	3.3456	0.31 h	22.1k tok/s	2.54 GiB
Medium · 50M	completed	50.0M	3.8072	3.1543	0.63 h	22.2k tok/s	2.54 GiB
High · 100M	completed	100.0M	3.3628	2.5812	1.26 h	22.2k tok/s	2.54 GiB